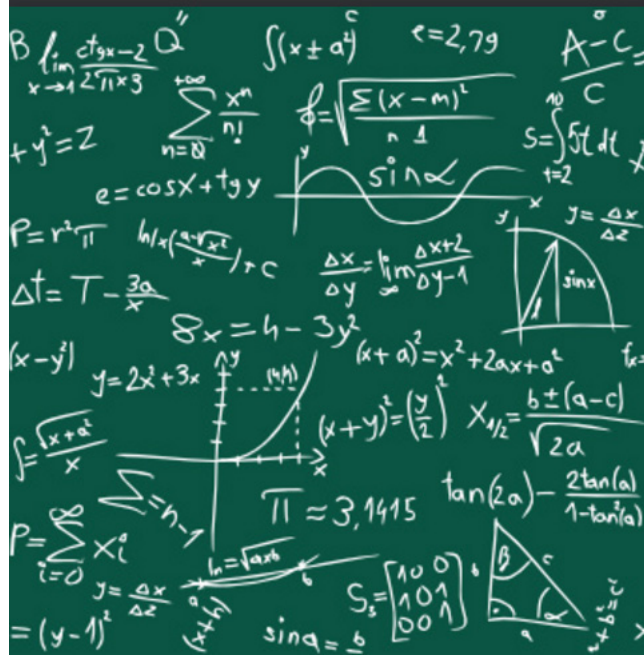


Vibration phenomena



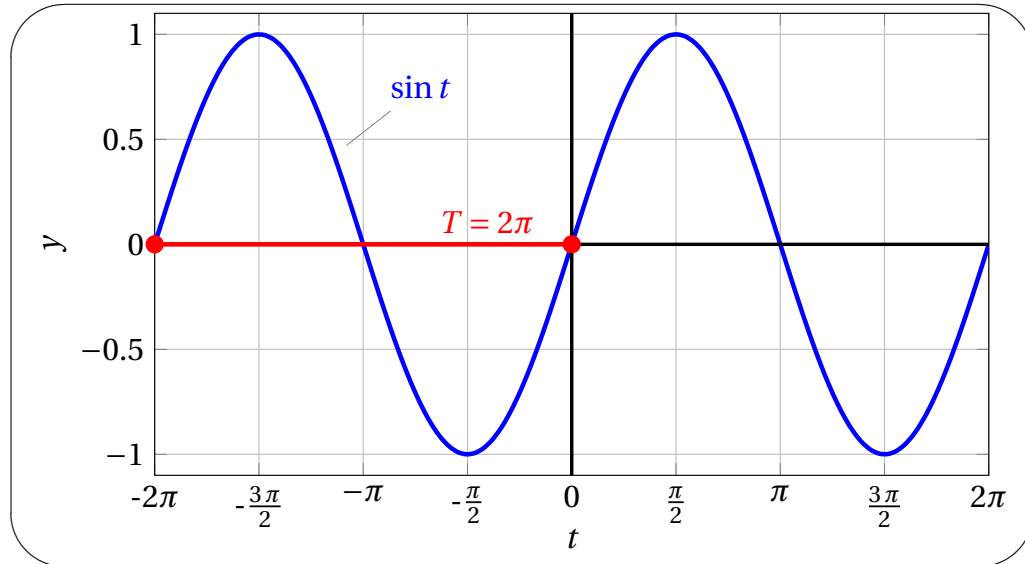
- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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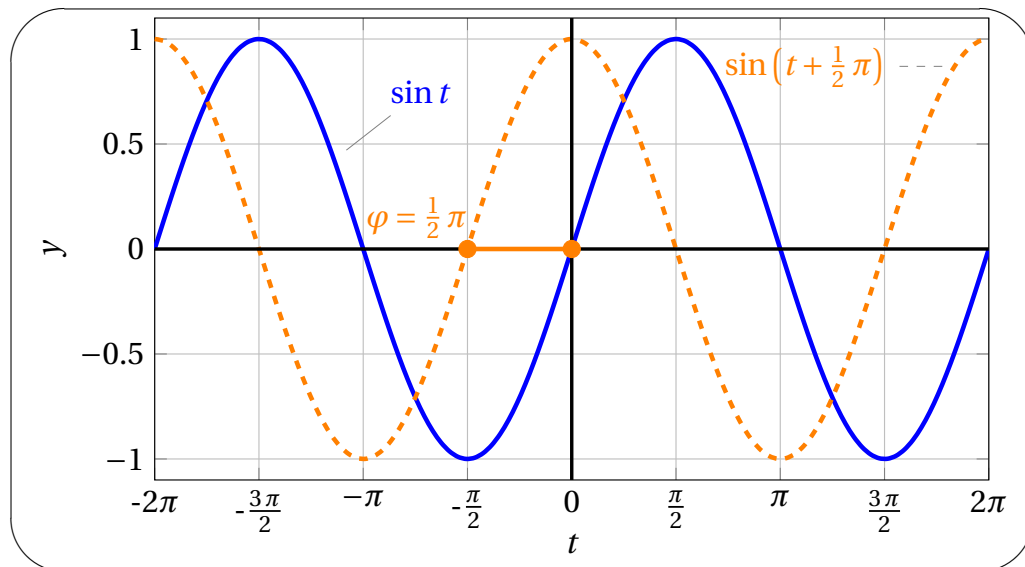
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1 Vibration phenomena

- We have seen that the sine and cosine functions are periodic with **period** 2π . We will use the variable t to denote time.



- Recall that $\sin(t + \varphi)$ corresponds to a **phase shift** φ (horizontal translation).



The functions:

$$t \mapsto a \sin(\omega t), \quad t \mapsto b \cos(\omega t) \quad (1)$$

where a , b and ω are positive real numbers, are called **elementary vibrations**. They are periodic functions with

$$\text{period } T = \frac{2\pi}{\omega}$$

Example 1: period

For example, considering $t' = t + \frac{2\pi}{\omega}$:

$$\sin \left[\omega \underbrace{\left(t + \frac{2\pi}{\omega} \right)}_{t'} \right] = \sin(\omega t + 2\pi) = \sin \omega t$$

- Clearly we have:

$$|a \sin(\omega t)| \leq a \quad \text{and} \quad |b \cos(\omega t)| \leq b$$

The functions (1) describe elementary vibrations characterized by

amplitude a and b (respectively)

$$\text{angular frequency } \omega = \frac{2\pi}{T}$$

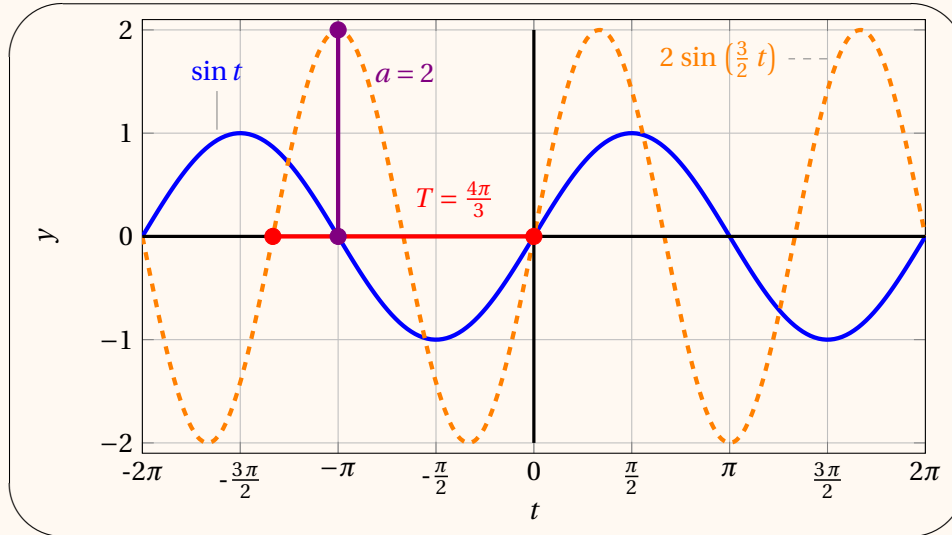
also called *angular velocity*, which indicates how many periods there are in an interval of length 2π . Moreover, they are characterized by the

$$\text{frequency } \nu = \frac{\omega}{2\pi}$$

which indicates how many times the function repeats itself in an interval of length 1

Example 2: of amplitude, angular frequency and frequency

$$t \mapsto 2 \sin\left(\frac{3}{2}t\right) \quad a = 2, \omega = \frac{3}{2}, T = \frac{4\pi}{3}$$



The angular frequency $\omega = \frac{3}{2}$ means that there are 1.5 periods in an interval of length 2π . The frequency ν is $\frac{3}{4\pi}$.

- The function

$$h(t) = a \sin(\omega t) + b \cos(\omega t) \quad (2)$$

describes the **superposition** of the two elementary vibrations with the **same angular frequency** ω . The latter is again an elementary vibration, **phase-shifted** with respect to the previous ones.

- Indeed, setting

$$A = \sqrt{a^2 + b^2}$$

we can write

$$h(t) = A \underbrace{\frac{a}{\sqrt{a^2 + b^2}}}_{\alpha} \sin(\omega t) + A \underbrace{\frac{b}{\sqrt{a^2 + b^2}}}_{\beta} \cos(\omega t) \quad (3)$$

- Now observe that the numbers

$$\alpha = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \beta = \frac{b}{\sqrt{a^2 + b^2}}$$

satisfy the conditions

$$-1 \leq \alpha \leq 1 \quad -1 \leq \beta \leq 1 \quad \alpha^2 + \beta^2 = 1$$

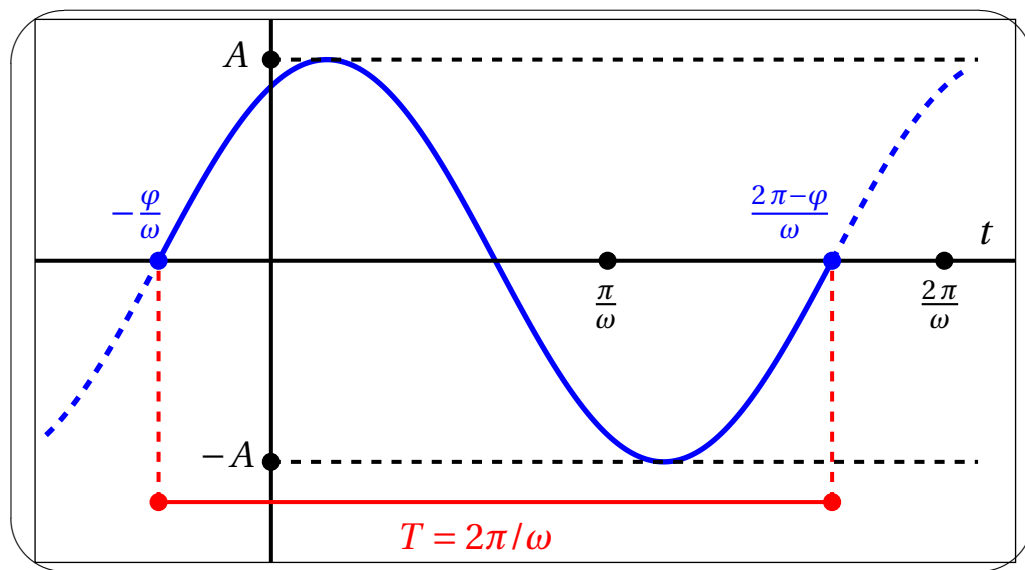
Hence there exists a **unique angle** φ such that

$$\cos \varphi = \alpha \quad \sin \varphi = \beta$$

so that (3) can be rewritten in the following form

$$\begin{aligned} h(t) &= A \cos \varphi \sin(\omega t) + A \sin \varphi \cos(\omega t) \\ &= A \sin(\omega t + \varphi) \end{aligned} \tag{4}$$

Therefore $h(t)$ represents an elementary vibration with amplitude A , angular frequency ω , phase-shifted by an angle φ .



In summary:

$$a \sin(\omega t) + b \cos(\omega t) = A \sin(\omega t + \varphi)$$

$$A = \sqrt{a^2 + b^2} \quad \begin{cases} a = A \cos \varphi \\ b = A \sin \varphi \end{cases}$$

- Under fairly general conditions, a periodic natural phenomenon can be written as the superposition of a finite or infinite number of elementary vibrations with different frequencies (**Fourier series**).

2 Damping and amplification effects

- By multiplying an elementary vibration by **powers** or **exponentials**, we can model **damping** or **amplification** effects.

First example

- For example, consider the function

$$h(t) = t \sin(\omega t)$$

which models an **amplified vibration**.

- Since $-1 \leq \sin(\omega t) \leq 1$, we have

$$-t \leq t \sin(\omega t) \leq t$$

and hence the graph of $h(t)$ lies between the graphs of the lines with equations $y = -t$, $y = t$.

- At the points where

$$\sin(\omega t) = 1$$

i.e.,

$$t = \frac{\pi}{2\omega} + k \frac{2\pi}{\omega} \quad (k = 0, 1, 2, \dots)$$

the graph of $h(t)$ touches that of $y = t$.

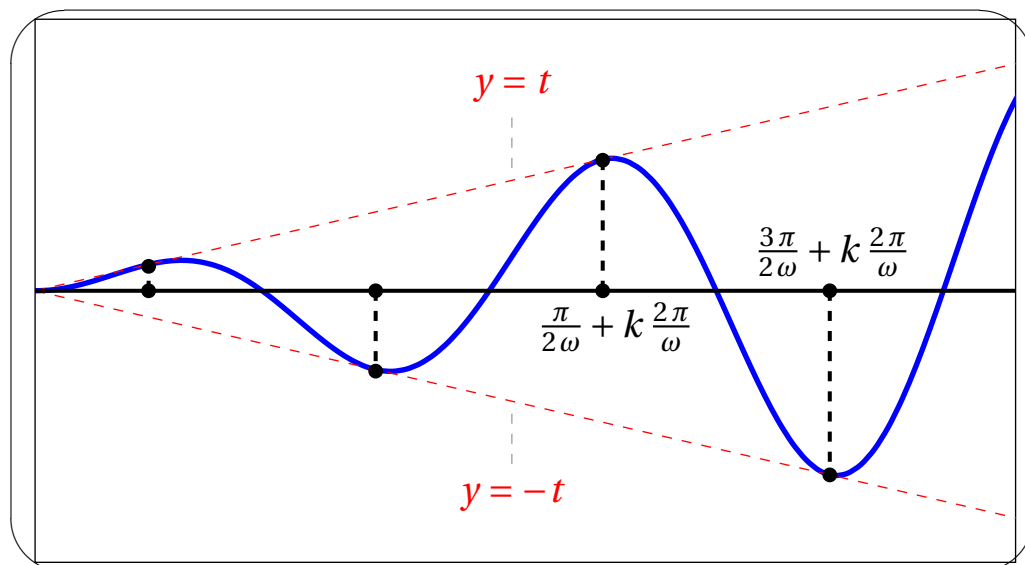
- At the points where

$$\sin(\omega t) = -1$$

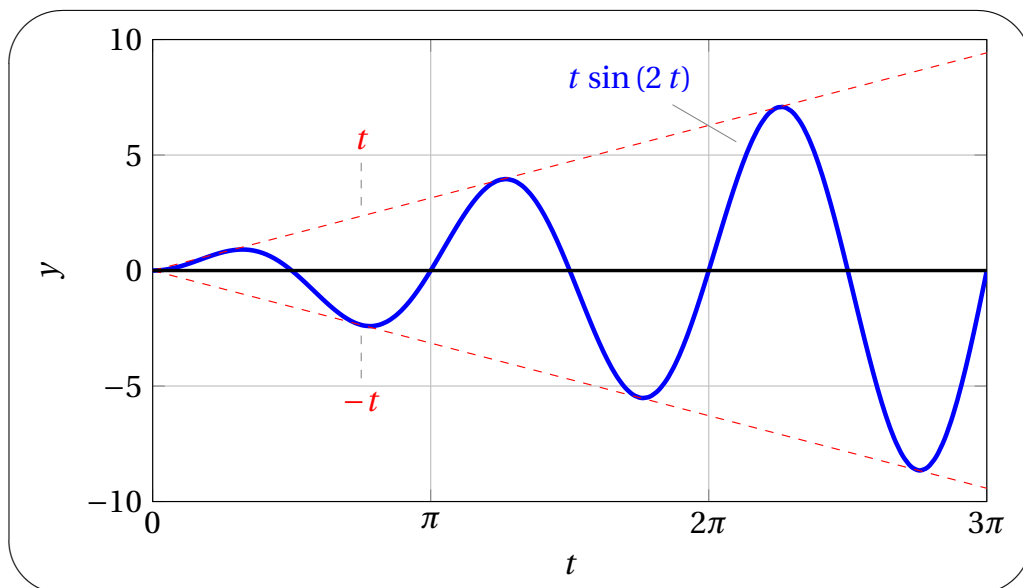
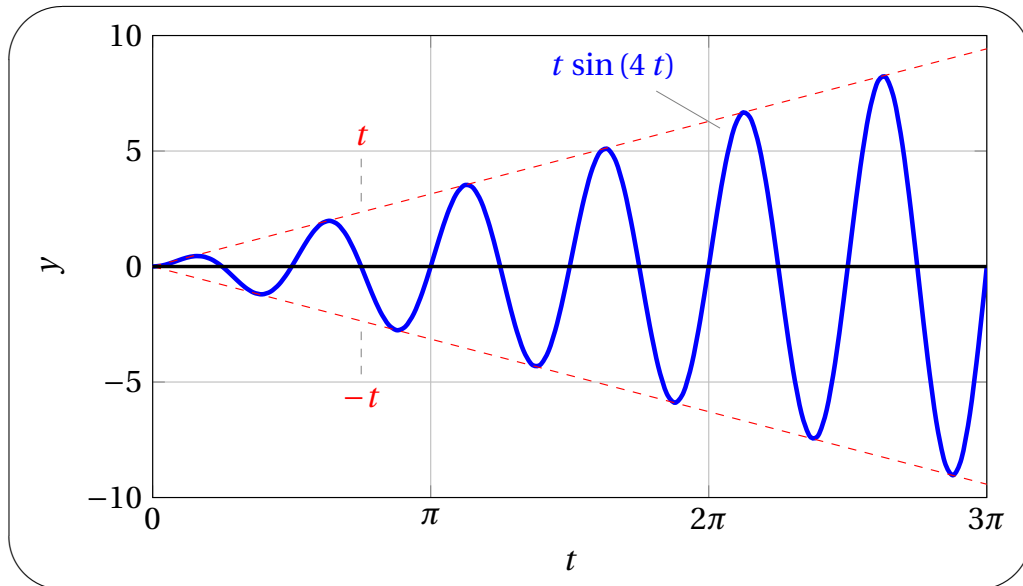
i.e.,

$$t = \frac{3\pi}{2\omega} + k \frac{2\pi}{\omega} \quad (k = 0, 1, 2, \dots)$$

the graph of $h(t)$ touches that of $y = -t$.



- From the graph we can see that multiplication by t has the effect of amplifying the vibration as t increases.



Second example

- For example, consider the function

$$k(t) = e^{-\alpha t} \sin(\omega t) \quad (\alpha > 0)$$

which models a **damped vibration**.

- Recalling that

$$e^{-\alpha t} = \left(\frac{1}{e^\alpha}\right)^t$$

is an exponential with base less than 1, considerations analogous to those made for the function $h(t)$ show that the graph of k lies between the graphs of the functions

$$y_1 = e^{-\alpha t} \quad \text{and} \quad y_2 = -e^{-\alpha t}$$

as shown in the figure:

